

ADOC Toolkit

Customer Guide

Acceldata Observability Cloud (ADOC) Toolkit — interactive CLI for API access and policy metrics export.

Version 0.1.0 · Customer-facing documentation

This guide covers installation, configuration, commands, export options, and **which CSV columns are applicable vs N/A** for each policy type.

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1. Overview & Prerequisites

The ADOC Toolkit connects to your Acceldata Observability Cloud instance to explore APIs, export rule-level execution metrics across multiple policy types, export rule/asset/alert summary metrics, and manage environments, HTTP, logging, and audit settings.

Prerequisites

Requirement	Notes
Python 3.10+	Required
uv (recommended) or pip	Dependency management
Network access to ADOC	Or offline install via S3
ADOC API credentials	access_key + secret_key
AWS CLI	Only for S3 offline package install

2. Installation

Standard install

```
git clone <repository-url>
cd adoc-toolkit
uv sync
uv sync --extra export # optional: Parquet / Avro
uv run adoc-toolkit
```

Launchers: Unix bin/adoc-toolkit · Windows bin\adoc-toolkit.bat

Offline / air-gapped (S3)

```
# Publish (machine with internet)
make s3-push S3_PATH=s3://your-bucket/adoc-toolkit/packages

# Install on target
make s3-pull S3_PATH=s3://your-bucket/adoc-toolkit/packages
```

3. Environment Configuration

Create config/environments.yaml before using API or export commands:

```
environments:
  production:
    name: "production"
    base_url: "https://your-instance.acceldata.app"
    access_key: "YOUR_ACCESS_KEY"
    secret_key: "YOUR_SECRET_KEY"
    timezone: "UTC" # Optional IANA name; NOT IST/EST

default_environment: "production"
```

Field	Required	Description
name	Yes	Must match YAML key
base_url	Yes	ADOC instance URL
access_key	Yes	API access key
secret_key	Yes	API secret key
timezone	No	IANA timezone (default UTC). Example: Asia/Kolkata

Timezone affects: export datetime headers (e.g. Started_At(Asia/Kolkata)) and tracking-file timestamps. Supports \${ADOC_ACCESS_KEY} / \${ADOC_SECRET_KEY}.

4. Command Reference

Select an environment with use before API/export commands.

Command	Aliases	Purpose
help	h, ?	List commands or detailed help
exit	quit, q	Exit the shell

Command	Aliases	Purpose
use	—	Switch environment
show-env	env	Show current environment (keys masked)
history	hist	Command / execution history
set-config	config, set	Configure HTTP, logging, audit
get	g	HTTP GET to ADOC APIs
find-asset	search, search-asset, asset-search	Search assets by name
export-metrics	export, metrics	Rule/asset/alert summary export
export-execution-metrics	exec-metrics, execution-metrics	Rule-level execution metrics
example	ex, demo	Developer template (internal)

Note: There is no separate show-config command. Use set-config --list or set-config --show <key>.

5. Toolkit Configuration (set-config)

Key	Default	Options
http.timeout	120	30, 60, 120, 300
http.retries	3	0, 1, 3, 5
http.proxy	null	URL, or none to clear
http.response.type	json	json, table, csv
log.level	TRACE	TRACE, DEBUG, INFO, ERROR
log.filepath	null	File path
log.rotate.onsize	10MB	10MB, 50MB, 100MB, 1GB
log.rotate.ontime	120	Minutes
audit.logfile	null	Audit log path

6. export-execution-metrics

Exports detailed rule-level performance metrics for operational monitoring, compliance, and analytics.

Supported policy types

CLI value	Notes
DATA_QUALITY	Default
EQUALITY	Reconciliation policies (default)
DATA_DRIFT	
PROFILE_ANOMALY	
SCHEMA_DRIFT	
FRESHNESS	Uses DATA_CADENCE APIs internally

Options

Flag	Default	Description
--output-type	csv	csv or parquet
--output-dir	./output/execution-metrics/	Output directory
--output-filename	execution-metrics-%d-%m-%y-%h-%M	Template; env name appended
--backload	none	-7d, -12h, 2024-01-15, ... (max 60 days)
--policy-types	DATA_QUALITY,EQUALITY	Comma-separated list
--page-size	100	API page size 1–1000

Incremental tracking

Scenario	Start time used
--backload provided	Backload value (tracking ignored)
No tracking file	30 days ago
Tracking file exists	last_run_timestamp from .last_run_tracking.json

Output files

Main consolidated file always written. Per-type files written when rows exist: data-quality-metrics, reconciliation-metrics, data-drift-metrics, freshness-metrics, schema-drift-metrics, profile-anomaly-metrics. Per-type files **drop columns that are entirely N/A**.

```
export-execution-metrics
export-execution-metrics --backload -7d
export-execution-metrics --policy-types DATA_QUALITY,EQUALITY,DATA_DRIFT,FRESHNESS
export-execution-metrics --output-type parquet --page-size 500
```

7. Column Applicability & N/A Rules

7.1 How N/A works

Situation	Export behavior
Column not applicable to a policy type	Literal "N/A" in consolidated CSV
Column applicable but value missing	Empty / null (not forced to N/A)
Null datetime	String "null"
Within EQUALITY only	Some fields use "NOT_APPLICABLE" (see 7.4)
Per-type CSV	Columns that are all N/A or all null are removed

7.2 Common / shared columns

Legend: **P** = Populated when data exists · **N/A** = Not applicable · **S** = Synthetic label

Column	DQ	EQ	DRIFT	FRESH	SCHEMA	PROFILE
Policy_Name / Policy_ID / Rule_Version	P	P	P	P	P	P
Execution_ID / Rule_ID	P	P	P	P	P	P

Column	DQ	EQ	DRIFT	FRESH	SCHEMA	PROFILE
Label_Key / Label_Value	P	P	S	S	S	N/A
Item_Measurement_Type	P	P*	P	P	N/A	P
Rule_Identifier	P	P	P	P	N/A	P
Rule_Success_Rate	P	P	P	P	N/A	N/A
Rows_Scanned	P	P	N/A	N/A	N/A	N/A
Rows_Failed	P	P	N/A	N/A	N/A	N/A
Total_Failed_Records	P	P	N/A	N/A	N/A	N/A
Rule_Result_Status	P	P	P	P	P	N/A
Overall_Policy_Status / Quality Score	P	P	P	P	P	P
Started_At / Finished_At / Execution_Date ({tz})	P	P	P	P	P	P
Execution_Status / Policy_Type / Policy_Enabled	P	P	P	P	P	P
Policy_Description	P	P	P	P	P	N/A
Rule_Description	P	P	N/A	N/A	N/A	N/A

* EQUALITY types: EQUALITY_MATCH or ROW_COUNT_MATCH. DQ=DATA_QUALITY, EQ=EQUALITY, DRIFT=DATA_DRIFT, FRESH=FRESHNESS, SCHEMA=SCHEMA_DRIFT, PROFILE=PROFILE_ANOMALY.

7.3 DATA_QUALITY-oriented columns

Column	DQ	EQ	DRIFT	FRESH	SCHEMA	PROFILE
Datasource_Name (derived)	P	N/A	P	P	P	P
Table_Asset_Name	P	N/A	P	P	P	P
Item_Column_Name	P	N/A	P	N/A	N/A	P
Rule_Strategy	P	N/A	P	P	N/A	N/A
Rule_Lower_Threshold	P	N/A	P	P	N/A	N/A
Rule_Upper_Threshold	P	N/A	P	P	N/A	N/A

7.4 EQUALITY / reconciliation columns

Column	DATA_QUALITY	EQUALITY	Other types
Left_Column / Right_Column	N/A	P (NOT_APPLICABLE for ROW_COUNT_MATCH)	N/A
Left_Rows_Scanned / Right_Rows_Scanned	N/A	P (NOT_APPLICABLE for EQUALITY_MATCH)	N/A
Use_For_Joining	N/A	P (NOT_APPLICABLE for ROW_COUNT_MATCH)	N/A
Operation / Join_Type	N/A	P	N/A
Left_Datasource_Name / Right_Datasource_Name	N/A	P (derived from asset UID)	N/A
Left_ASSET_UID / Right_ASSET_UID	N/A	P	N/A

In reconciliation-only files, Rows_Failed may be renamed to Rows_Failed/Drift when ROW_COUNT_MATCH rows are present.

7.5 Policy-type-specific columns

Column	Applicable only for	All other types
Drift_Threshold	DATA_DRIFT	N/A
Anomaly_Detected	FRESHNESS	N/A
Threshold_Breached	FRESHNESS	N/A
Metric_Anomalous	PROFILE_ANOMALY	N/A
Asset_Addition	SCHEMA_DRIFT	N/A
Asset_Deletion	SCHEMA_DRIFT	N/A
Data_Type	SCHEMA_DRIFT	N/A
Asset_Relation_Change	SCHEMA_DRIFT	N/A
Asset_Metadata	SCHEMA_DRIFT	N/A

Column	Applicable only for	All other types
Metadata_Configs	SCHEMA_DRIFT	N/A

7.6 Derived columns

Column	How derived	When N/A
Datasource_Name	First segment before '.' in Table_Asset_Name	EQUALITY
Left_Datasource_Name	First segment of Left_ASSET_UID	Non-EQUALITY
Right_Datasource_Name	First segment of Right_ASSET_UID	Non-EQUALITY
Rule_Identifier	Policy item / mapping / display name	SCHEMA_DRIFT
Total_Failed_Records	Execution failedRows	Drift / Freshness / Schema / Profile
Metric_Anomalous	isMetricAnomalous from profile anomaly details	All except PROFILE_ANOMALY

7.7 Labels vs Tags

Concept	Command	Columns	Source
Labels	export-execution-metrics	Label_Key, Label_Value	Policy detail labels (1 row per label)
Tags	export-metrics	Tags	rule.tags[].name only (names, not IDs)

Synthetic labels: DATA_DRIFT = {columnName}-{metricType}; FRESHNESS = {measurementType}-{strategy};
 SCHEMA_DRIFT = Policy_Name / item id.

8. export-metrics

Exports enabled/active rules with latest execution metrics, catalog asset attributes, and related alerts.

Flag	Default	Description
--output-type	csv	csv, parquet, or avro
--output-dir	./output/export-metrics/	Output directory
--output-filename	ad-metrics-%d-%m-%y-%h-%M	Template + env suffix
--page-size	100	1–1000

Columns

Group	Columns
Always	Rule Name, Rule ID, Rule Type, Asset ID, Execution Status, Execution Date ({tz}), Quality Score, Records Processed, Execution Duration (ms), Open Alerts, Tags
When asset found	Asset Name, Asset UID, Source Type, Asset Type, Asset Quality Score, Open Alert Count, Open Alert ID, Open Alert States
When alert found	Alert ID, Alert Total Count, Alert Created/Updated At ({tz}), Alert Status, Assignee, Updated By, Severity

Tags are tag **names** only (not IDs), e.g. {'critical', 'prod'}, or N/A if none. Missing scalars → N/A; missing timestamps → "null".

9. APIs Used

export-execution-metrics

GET /catalog-server/api/rules/executions · Result APIs under data-quality | reconciliation | data-drift | profile-anomaly | schema-drift | data-cadence · Policy detail APIs · Asset overview & search.

export-metrics

GET /catalog-server/api/assets/list · GET

/catalog-server/api/rules?withLatestExecution=true&ruleStatus=ENABLED,ACTIVE · Incidents listing for alerts.

Auth headers: accessKey, secretKey from the active environment.

10. Workflows & Troubleshooting

Recommended workflows

```
# Daily incremental
use production
export-execution-metrics

# Weekly historical refresh
export-execution-metrics --backload -7d \
  --policy-types DATA_QUALITY,EQUALITY,DATA_DRIFT,FRESHNESS,SCHEMA_DRIFT,PROFILE_ANOMALY

# Summary metrics
export-metrics --output-type csv --page-size 500
```

Troubleshooting

Issue	What to check
No environment selected	Run use &environment&
Empty export	Broaden --backload; check --policy-types; verify policies ran
Many N/A columns	Expected in consolidated CSV; use per-type files for cleaner schemas
Parquet/Avro fails	uv sync --extra export
Auth / 401	Credentials in environments.yaml
Timezone wrong	Use IANA names (Asia/Kolkata), not IST
Offline install	AWS CLI credentials; correct S3_PATH

Generated for customer use from the ADOC Toolkit codebase. Column applicability reflects the current export implementation.